AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 32 · STUDENT LESSON

Project Problem and Success Test

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Weeks 2 and 23: requirements and scores are separate.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Define a small application purpose. Try the recall before opening notes.

Your learning target

I can define a small application purpose.

I can turn **requirement**s into checkable **acceptance case**s.

I can plan a feasible **pipeline** and human review.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Permissions and Human Review. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from **Evidence** and Sources. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Image and Voice **Pipeline**s. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 32: Project Problem and Success Test

A project is defined by **user need** and measurable success, not feature count.



THINKER MISSION

Write a recommender brief and three **acceptance test**s.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Why must a recommendation show no eligible option when every item fails a hard constraint?

Readiness for A reproducible data investigation

What should happen to an invalid duration instead of silently making it zero?

My readiness explanation

This week's end goal

By the end, I can plan a recommender with **acceptance test**s before writing its selection code.

Notice the evidence

> Investigate the supplied Project Problem and Success Test case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use project problem and success test to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Plan a fictional activity chooser for a teacher-supervised 20-minute session. It will validate a small local dataset, exclude over-limit items, rank eligible items using 2 * fun + learning and break ties alphabetically.

It will explain the result for human review. It will not book, message or collect personal student profiles.

Apply the connected idea: A reproducible data investigation

Investigate the fictional question: among supplied sessions with valid duration records, what is the mean duration? **Input**s are 10, 0, 20, None and -2 minutes. Our schema accepts exact integers 0 through 60.

The wording deliberately limits the answer to accepted records.

STUDENT LESSON · 3 OF 18

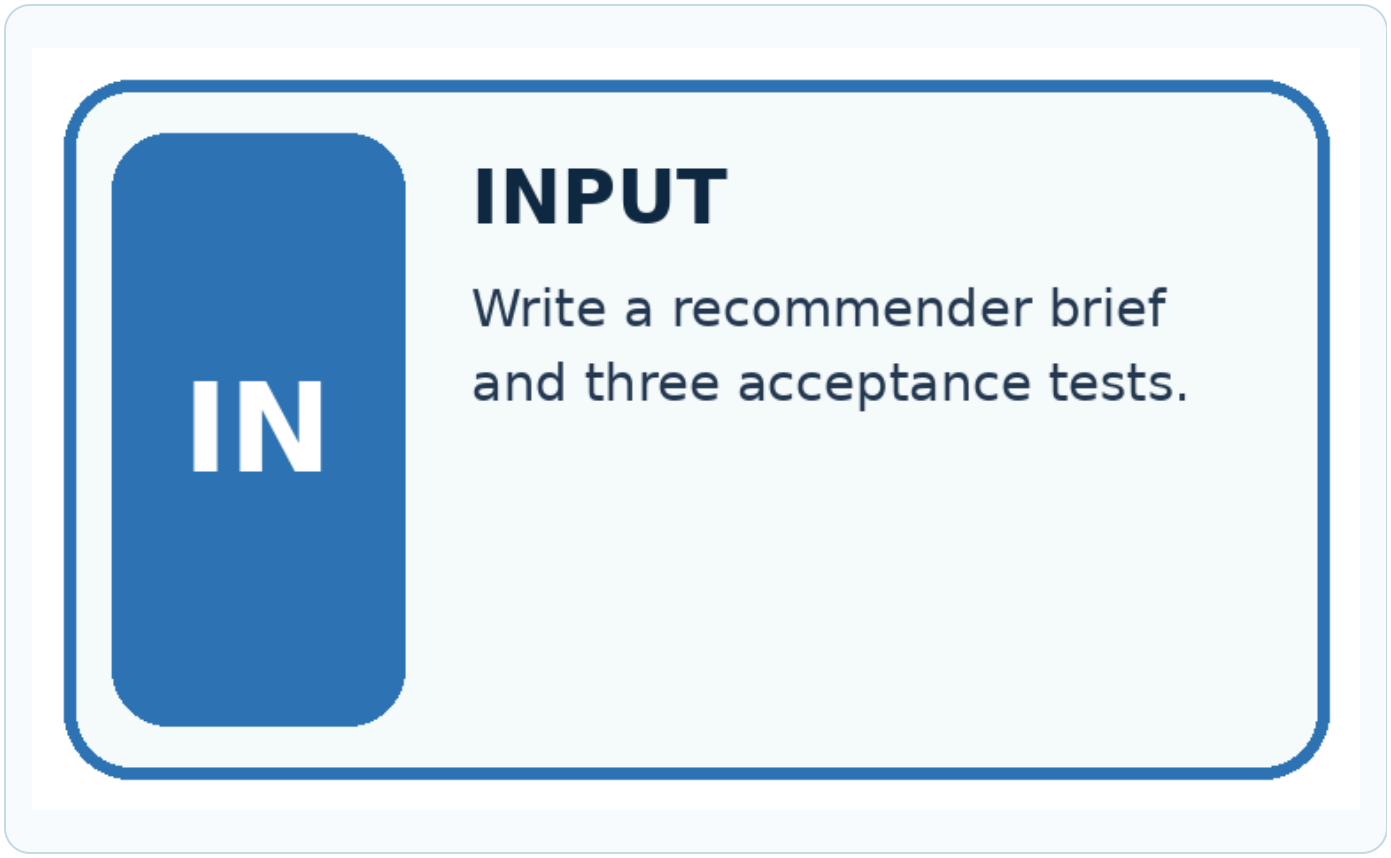
See how it works

> Follow the mechanism for Project Problem and Success Test. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Write a recommender brief and three acceptance tests.

↓


PROCESS
A project is defined by user need and measurable success, not feature count.

↓



Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

```
requirements = {"limit": 20, "fun_weight": 2,
                "learning_weight": 1, "tie": "name ascending"}
print(requirements["limit"])
print(requirements["tie"])
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Input	Local fictional records	Validate schema
Decision	Filter then score then tie rule	Explain selected item
Output	Suggestion or no eligible item	Teacher reviews

Read each row from left to right and connect the evidence to the stated rule.

Define fields name, minutes, fun and learning. Require a nonempty text name, integer minutes 0 through 60 and integer ratings 0 through 5. Reject duplicate names because the tie rule uses names as identifiers.

Write cases before implementing so **requirement**s do not drift to fit **output**s.

Turn a wish into a checkable plan

A phrase such as make a smart choice does not say what counts as success.

The chooser plan does: validate the named fields, include durations at most 20 minutes, calculate 2 times fun plus learning, and use alphabetical names to break a tie.

Describe the expected **output** and a no-match result before coding. Another learner should be able to use your plan to predict the same answer without guessing what you intended.

Apply the connected idea: A reproducible data investigation

```
raw = [10, 0, 20, None, -2]
valid = []
rejected = 0
for value in raw:
```

```
    if type(value) is int and 0 <= value <= 60:
        valid.append(value)
    else:
        rejected = rejected + 1
print(len(valid), rejected)
if len(valid) > 0:
    print(sum(valid) / len(valid))
else:
    print("No valid observations")
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Raw evidence	Five supplied values	Keep unchanged
Validation	Accept 10, 0, 20	Reject unknown and negative
Summary	30 divided by 3	Mean 10 minutes

Read each row from left to right and connect the evidence to the stated rule.

The result is 3 accepted, 2 rejected, and mean 10.0 minutes. A reproducible report states the question, raw **input** version, schema, exclusion reasons, calculation and **scope**.

Someone else should obtain the same result from the same **input**s and rules.

Keep the denominator with the evidence

First filter the raw durations using the written schema. Keep 10, 0 and 20; reject None because the duration is unknown and -2 because it is outside the range.

The accepted sum is 30 and the accepted count is three, so the accepted-record mean is 10 minutes. Dividing by all five supplied records would treat the two rejected durations as if they contributed known zeros.

State three accepted out of five beside the mean.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

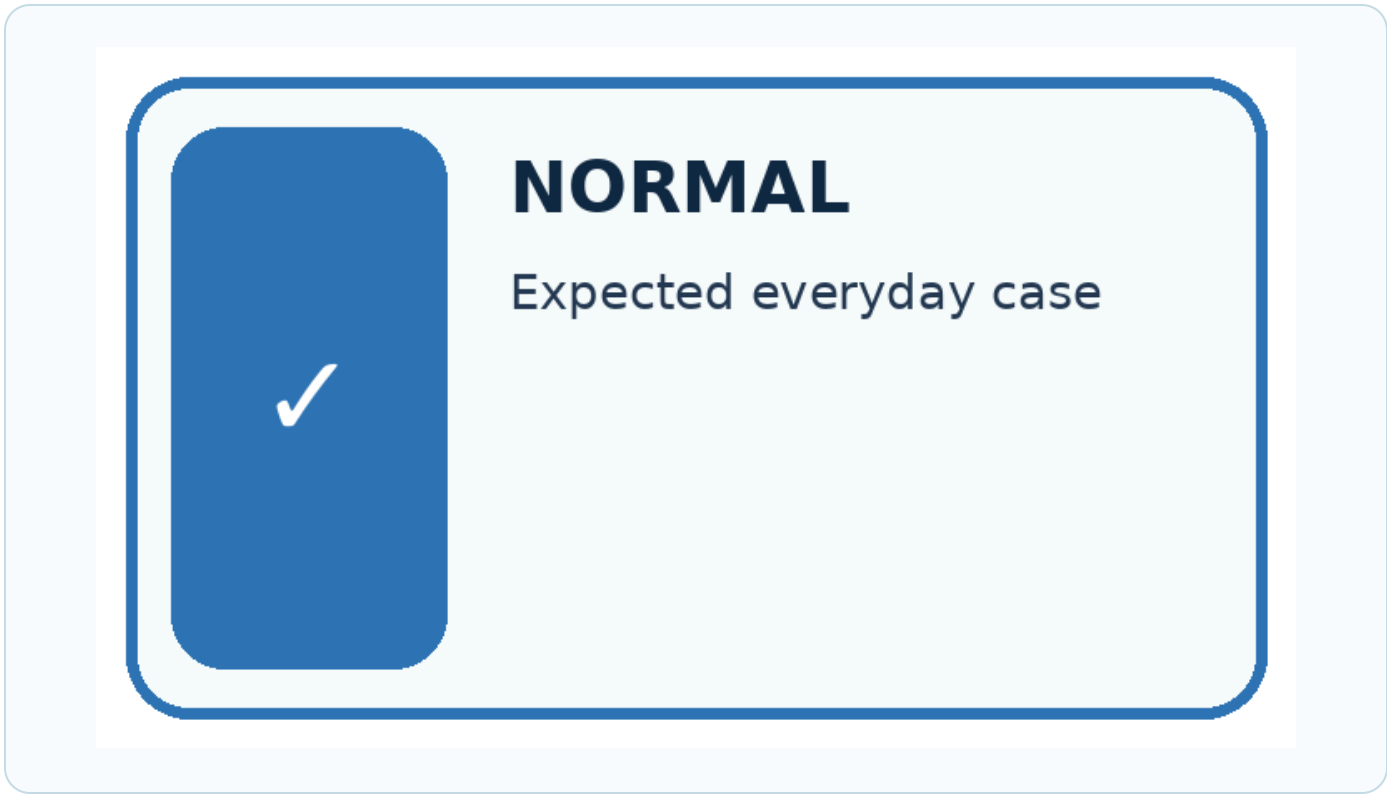
> Compare the supplied cases for Project Problem and Success Test. Identify the detail that changes the conclusion.

The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

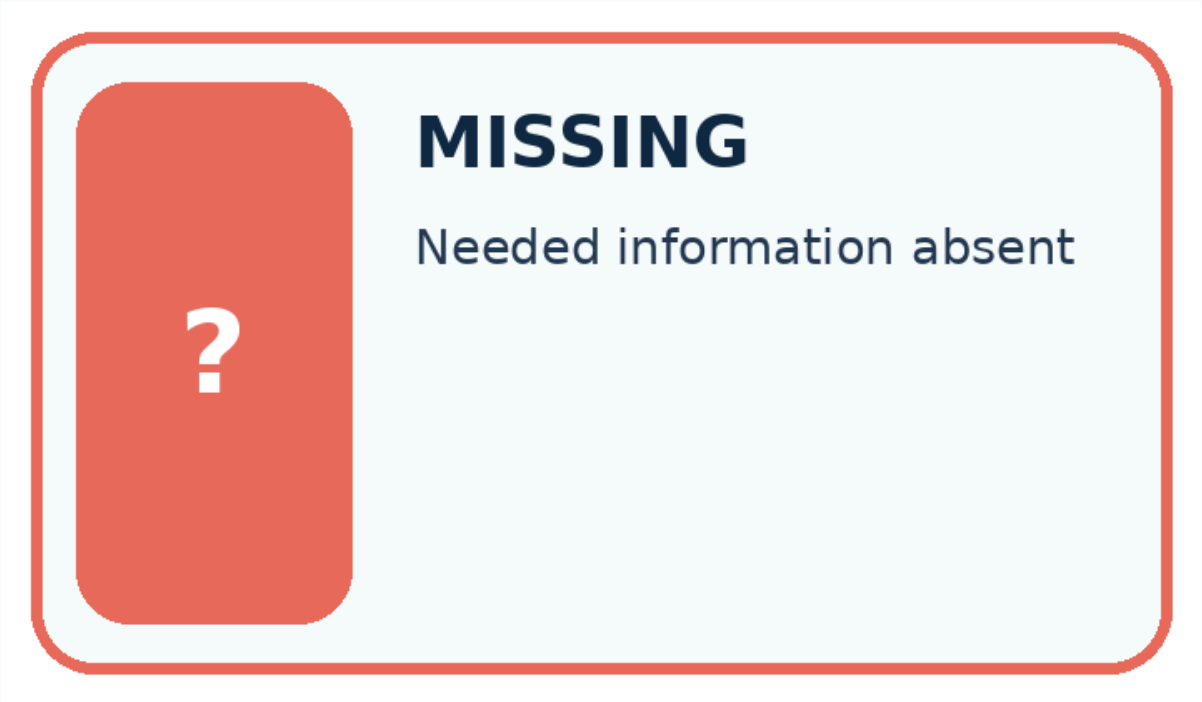
A system is not understood until it survives more than one comfortable example.


NORMAL
Expected everyday case


BOUNDARY
Near a decision edge

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Reduce the limit from 8 to 5.

Predicted result

Prediction: Return no eligible item.

Why the result follows

Reason: All three valid rows take more than five minutes.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Project Problem and Success Test to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
USER NEED
specific problem for a specific person


2
REQUIREMENT
behavior the solution must provide





Word	My own example	Not this / boundary
user need	The problem or desired outcome the system is meant to address for a person.	
requirement	A behaviour or condition the proposed system must satisfy.	

Word	My own example	Not this / boundary
acceptance test	A predefined check that a product satisfies a stated requirement.	
scope	The tasks, users, information and actions included within an agreement or design.	

Words used in the closer explanation

Word	Meaning
Acceptance case	An input and expected behavior used to judge a requirement
Non-goal	Something intentionally outside project scope
Prototype	A limited implementation for testing a design

Terms for A reproducible data investigation

Term	Meaning
Investigation	A documented process for answering a question with evidence
Pipeline	An ordered sequence of processing steps
Audit trail	A record of inputs, decisions and changes
Limitation	A reason a conclusion has restricted scope

STUDENT LESSON · 6 OF 18

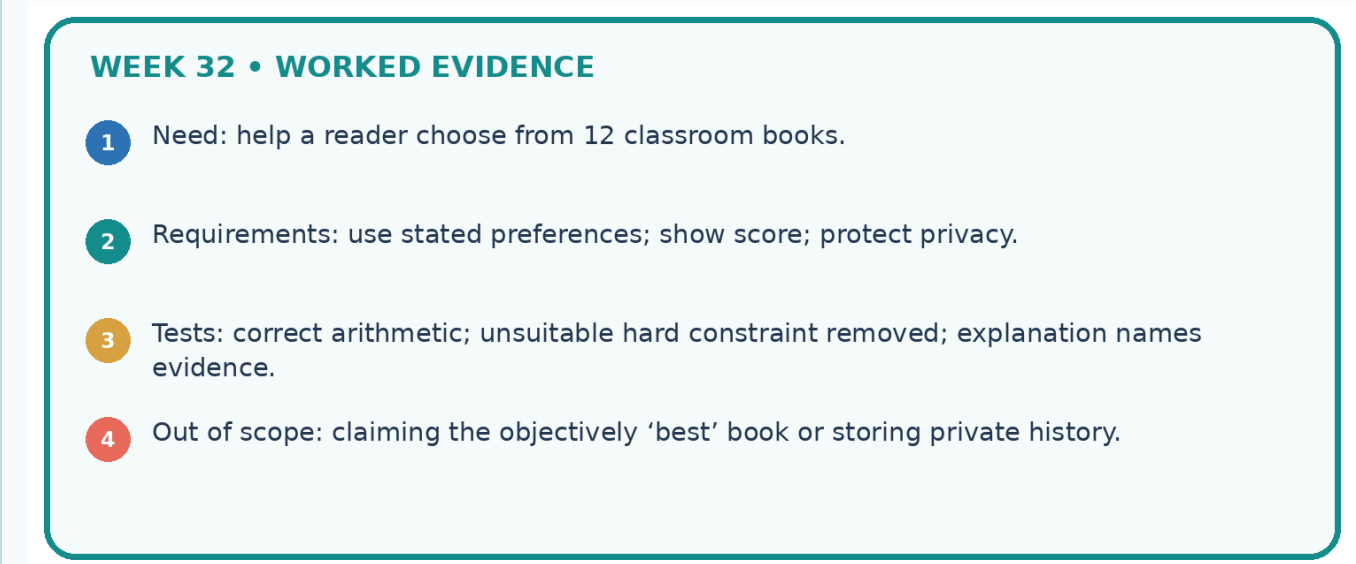
Follow a worked example

> Follow the worked trace for Project Problem and Success Test, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Need: help a reader choose from 12 classroom books.

2. **Requirement**s: use stated preferences; show score; protect privacy.

3. Tests: correct arithmetic; unsuitable hard constraint removed; explanation names **evidence**.

4. Out of **scope**: claiming the objectively ‘best’ book or storing private history.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Write an acceptance case for the duration boundary and one for no eligible option.

Apply the connected idea: A reproducible data investigation

If another valid value 30 is added, calculate the new accepted count and mean.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Project Problem and Success Test

A fictional reading-break chooser needs a transparent contract.

The facts we will use

Supplied facts

Items have id text, minutes integer 0–30, interest integer 0–5. Request limit is integer 0–30.

Supplied facts

Eligible if minutes<=limit; score=interest; highest wins; ties choose alphabetically smaller ID; invalid rows excluded and counted.

AI ACADEMY / AI-2/32

Project Problem and Success Test

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Specify inputs

Validate each row and request; keep rejected-row reasons.

2. Filter and rank

At limit 8: A(time8,interest3), B(9,5), C(6,3). B is ineligible.

3. Expected output

A beats tied C by ID; report A and its score 3.

4. Acceptance cases

Also plan invalid request, no eligible item and equal-score cases.

CHANGE ONE THING

Reduce the limit from 8 to 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Specify inputs	Validate each row and request; keep rejected-row reasons.
2. Filter and rank	At limit 8: A(time8,interest3), B(9,5), C(6,3). B is ineligible.
3. Expected output	A beats tied C by ID; report A and its score 3.
4. Acceptance cases	Also plan invalid request, no eligible item and equal-score cases.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Planning fixes what success means before code returns an answer. The duration boundary is including equality, and the tie policy prevents accidental list-order dependence.

A complete **output** may include why other rows were excluded. Test expectations should come from the contract rather than copying the first program result.

What this example does not establish

Keep the stated limit

The supplied interest scores are fictional; the specification does not establish genuine learner preferences.

Keep the stated limit

Read the labelled picture: Write an equality test and explain which **requirement** it checks.

Keep the stated limit

Change one thing: Reduce the limit from 8 to 5. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Project Problem and Success Test claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A project is better when it has more features. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

The **scope** is a rule-based **prototype** using fictional ratings. It does not demonstrate personalized learning impact or a trained AI model. A clear **non-goal** protects students from building an untestable or overly broad project.

> **Name two capabilities excluded from this prototype and explain why.**

Apply the connected idea: A reproducible data investigation

Missing or invalid durations might belong to unusually long sessions. The accepted mean may therefore differ from the full group's mean. Do not make a causal claim from this descriptive **investigation**.

> **A report gives only mean 9. What additional evidence lets a peer reproduce it?**

STUDENT LESSON · 8 OF 18

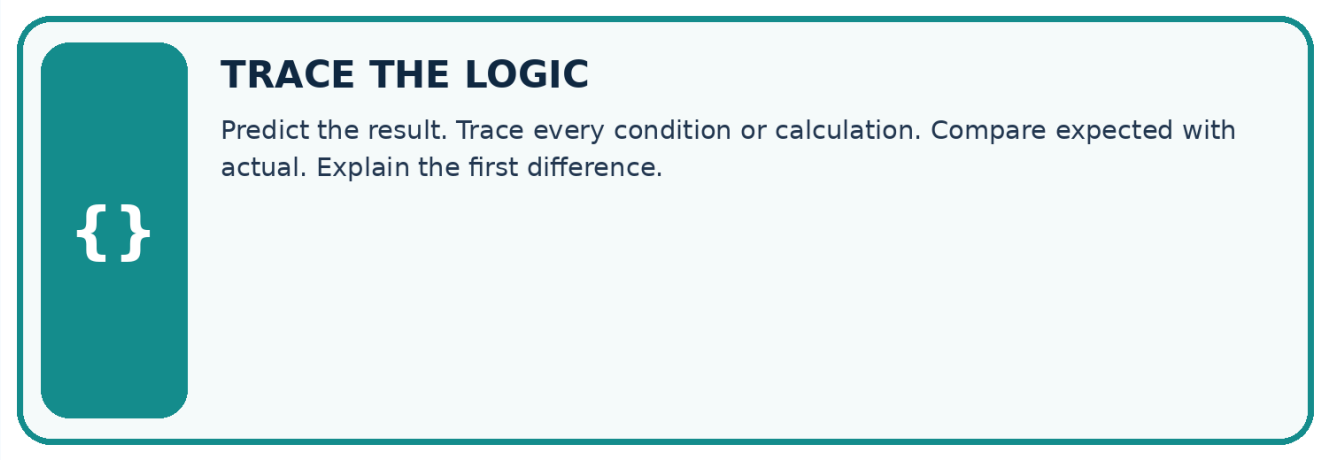
Practise together

> Practise Project Problem and Success Test with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, **limitation**, and next safe step

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

A **requirement** such as make a smart choice cannot be tested precisely. Replace it with a stated score, limit, tie rule and **output** explanation.

If real feedback later changes the policy, record a versioned change rather than rewriting previous **evidence**.

Apply the connected idea: A reproducible data investigation

Dividing the accepted total 30 by all five records gives 6 and answers no clear observed-duration question.

Use the accepted count for this mean, disclose exclusions, and never describe it as the average for every supplied session.

Practise with less help: Write an acceptance case

Use workbook W2 for the connected example “Planning the application”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Planning the application. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: A reproducible data investigation

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

Explain the picture

Write an equality test and explain which **requirement** it checks.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

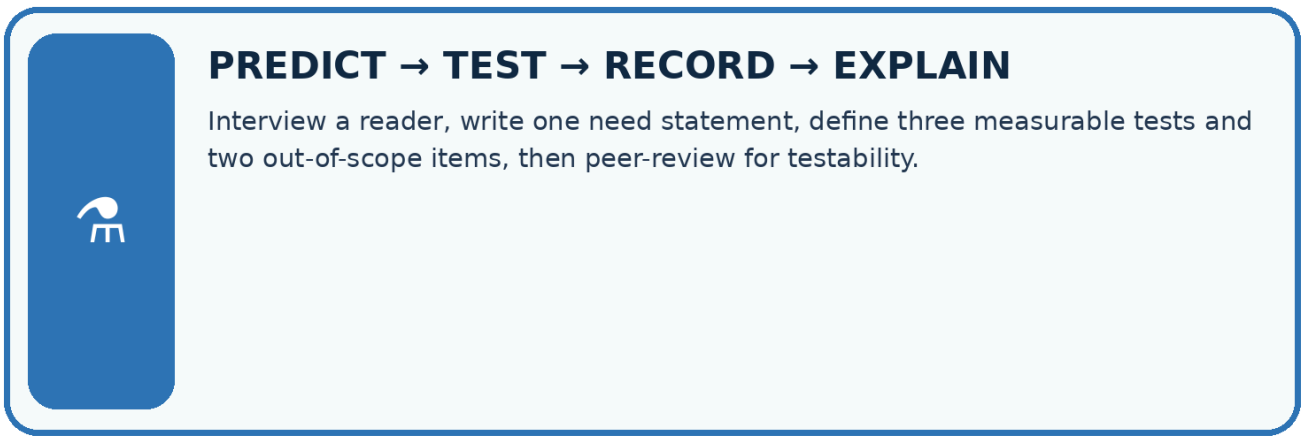
Run the investigation

> Investigate Project Problem and Success Test. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Interview a reader, write one need statement, define three measurable tests and two out-of-**scope** items, then peer-review for testability.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

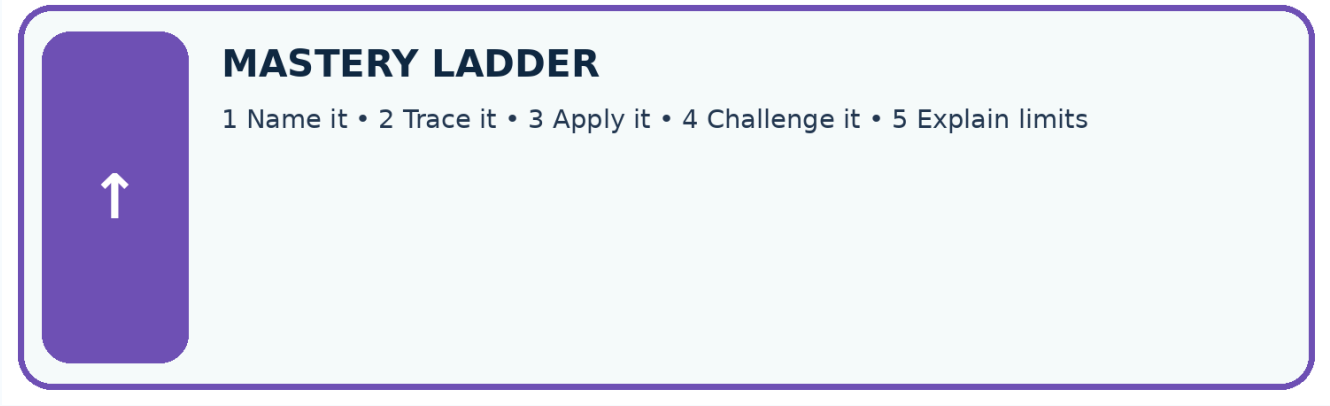
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Project Problem and Success Test. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

STUDENT LESSON · 11 OF 18

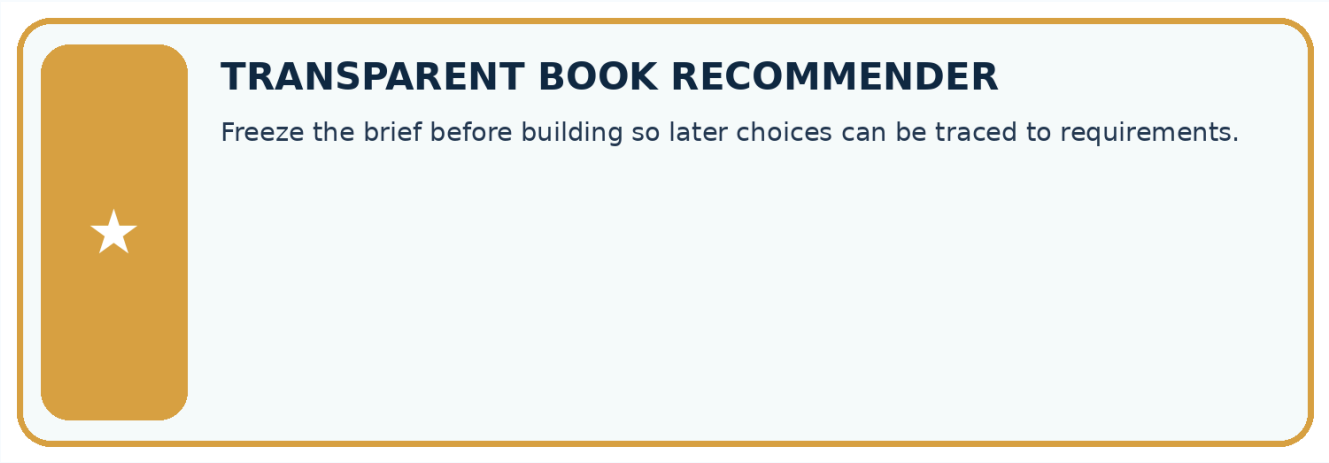
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Freeze the brief before building so later choices can be traced to **requirement**s.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Retain the Transparent Book Recommender's schema, constraints, score, tie rule and acceptance table as its build specification.

Show what you know

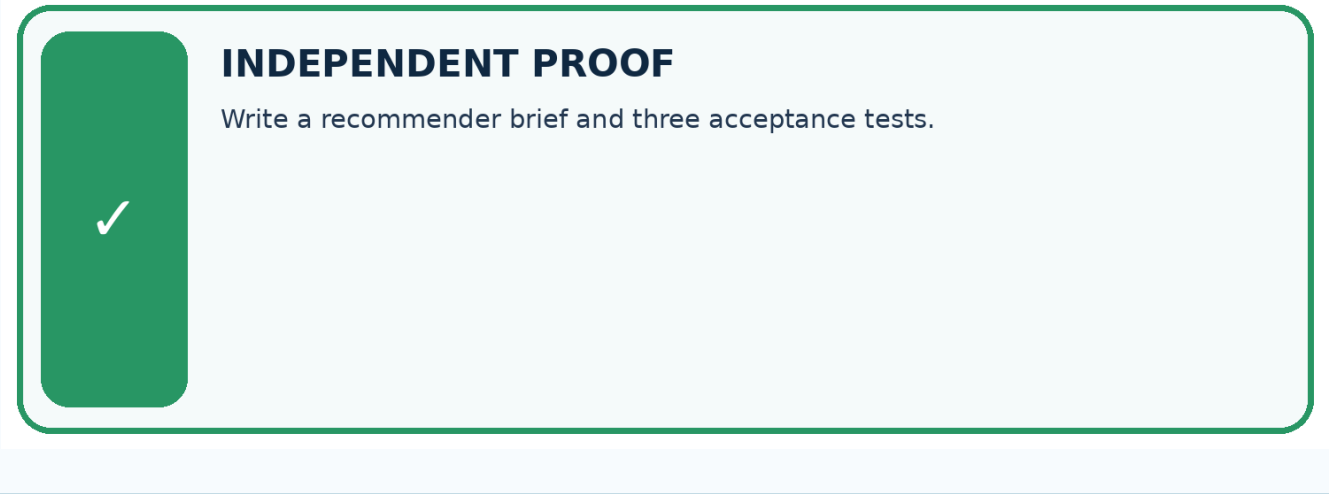
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Write a recommender brief and three **acceptance test**s.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One **limitation**, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a **limitation** ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Project Problem and Success Test case for supported practice. Repeated examples do not become fresh assessments.

Week 32 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A project recommends from ten approved books using genre and length.

The question

Define **acceptance test**s.

Worked explanation

Verify book existence, hard constraints, correct scoring explanations, tie handling and no-match behaviour on predefined requests.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Turn a project goal into acceptance tests

Evidence or stage	Value or result
Catalogue	Suggestion must be one of ten approved books
Constraints	Genre and length respected
Scoring	Explanation matches the calculation
Boundaries	Handle ties and no matches

A project recommends from ten approved books using genre and length.

Define **acceptance test**s.

Verify book existence, hard constraints, correct scoring explanations, tie handling and no-match behaviour on predefined requests.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Plan independently

Plan a 15-minute fictional club chooser. Specify schema, score, duration limit, tie rule, invalid-data behavior, **output** and three **acceptance case**s.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 32 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Define a testable goal for a book recommender serving readers who want short space books. Include a concrete success condition.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Freeze the brief before building so later choices can be traced to **requirement**s.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Specify three **acceptance test**s for a recommender: a normal match, a tied score and no qualifying book. Include expected behaviour before implementation.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Project Problem and Success Test, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional reading-break chooser needs a transparent contract.

By the end, I can	How I will show it
Define a small application purpose	Explain the deciding step, show a trace or test, and state one limit.
Turn requirements into checkable acceptance cases	Explain the deciding step, show a trace or test, and state one limit.
Plan a feasible pipeline and human review	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Project Problem and Success Test without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Define a small application purpose

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional reading-break chooser needs a transparent contract.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Also plan invalid request, no eligible item and equal-score cases.

1. Filter and rank

2. Specify **input**s

3. **Acceptance case**s

4. Expected **output**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: **Acceptance case**s Also plan invalid request, no eligible item and equal-score cases. Planning fixes what success means before code returns an answer.

The duration boundary is including equality, and the tie policy prevents accidental list-order dependence. A complete **output** may include why other rows were excluded.

Test expectations should come from the contract rather than copying the first program result.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Reduce the limit from 8 to 5. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Return no eligible item.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Return no eligible item. All three valid rows take more than five minutes.

Repair a missing building block

Review Permissions and Human Review (ai-2/31). Goal: Explain the **mechanism** and terminology of least privilege and tool **scope**s. Short practice: Which proposal is allowed, and why is an authorised object alone insufficient?.

Return to this week and explain what you changed.

Review **Evidence** and Sources (ai-2/29). Goal: Explain the **mechanism** and terminology of claim support and citation evaluation.

Short practice: Why is the original answer not fully supported despite having citations on every claim?. Return to this week and explain what you changed.

Review Image and Voice **Pipeline**s (ai-2/26). Goal: Explain the **mechanism** and terminology of speech **pipeline**s and multimodal **evidence**.

Short practice: Which stage first fails, and why should the faulty transcript not be silently overwritten?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

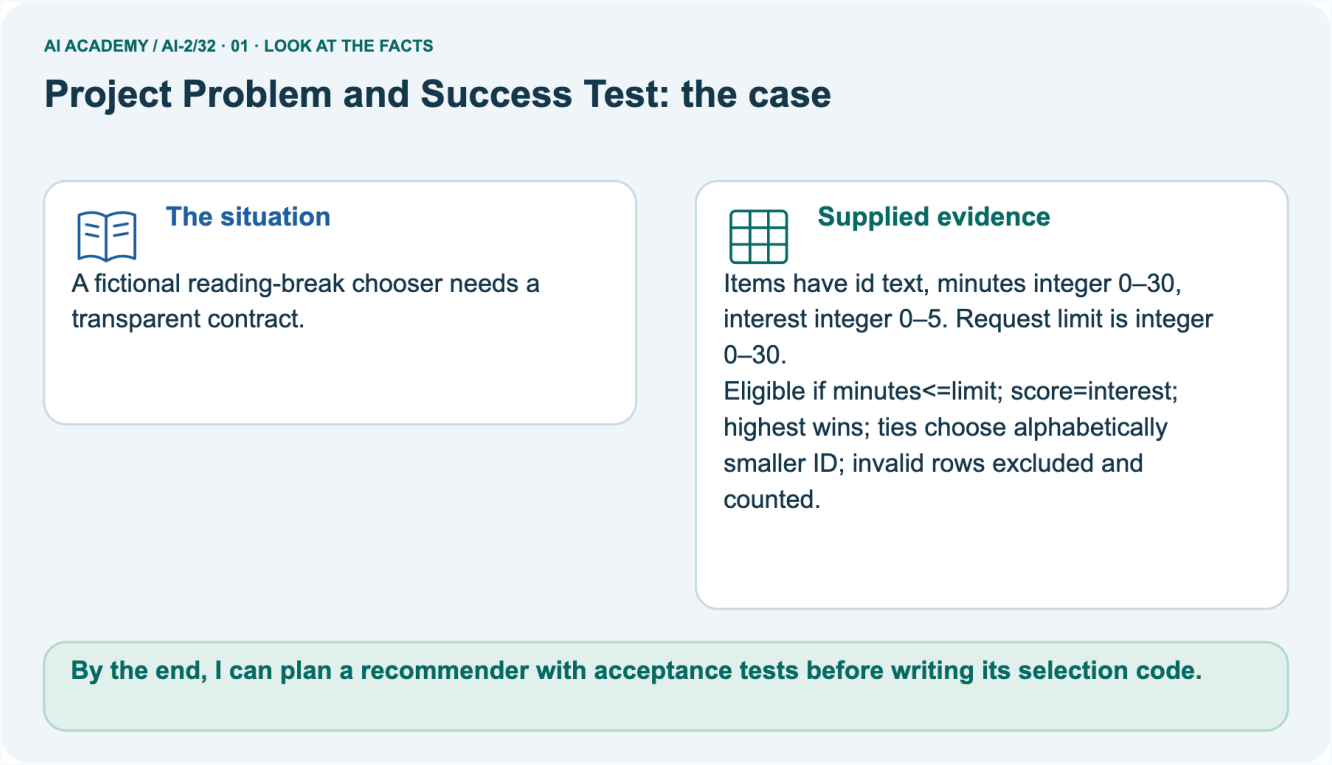
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Project Problem and Success Test graphic using its labels, deciding evidence and relevant limitation.

01. Project Problem and Success Test: the case



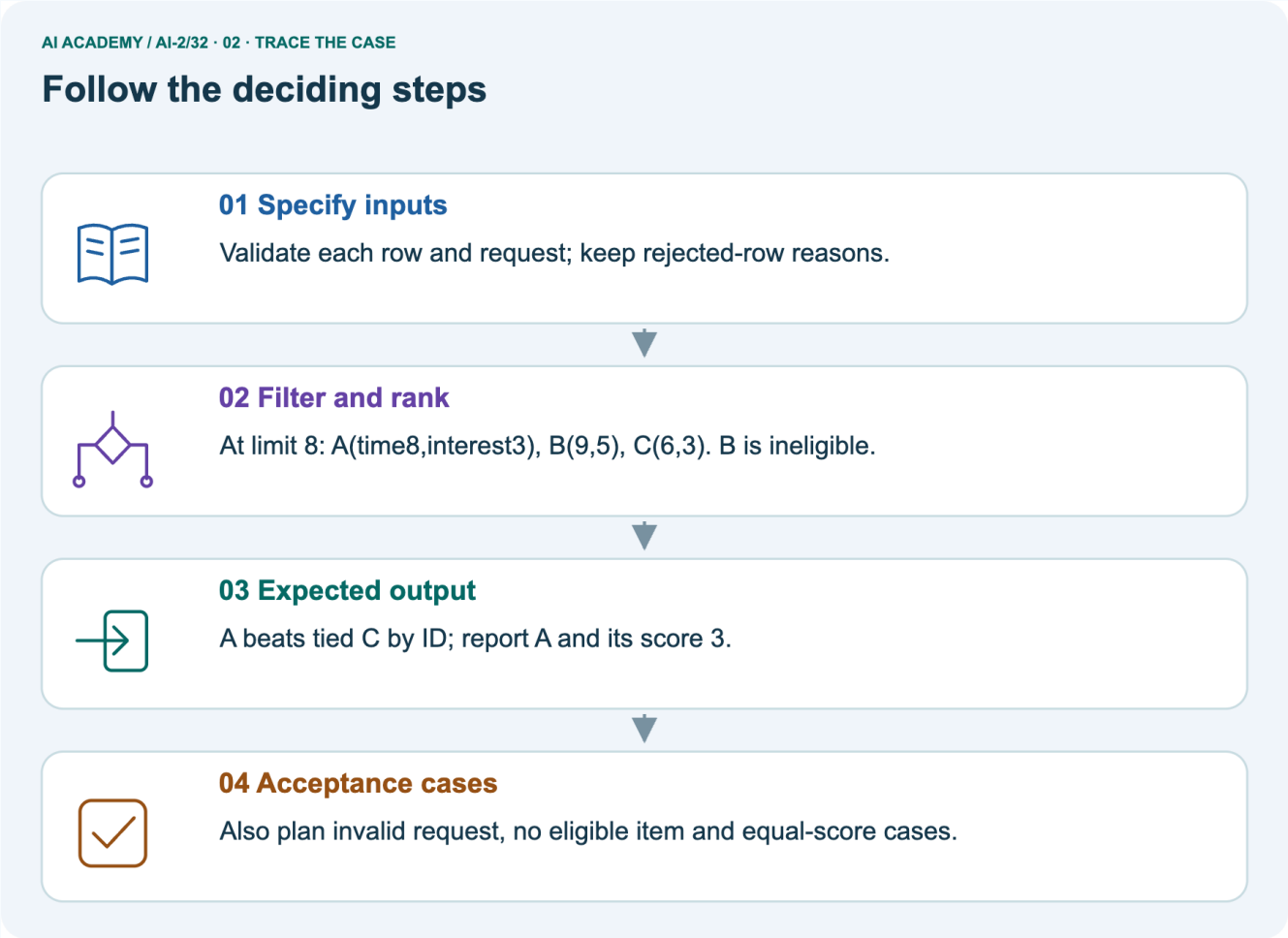
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional reading-break chooser needs a transparent contract. Supplied facts: Items have id text, minutes integer 0–30, interest integer 0–5. Request limit is integer 0–30.

Eligible if minutes<=limit; score=interest; highest wins; ties choose alphabetically smaller ID; invalid rows excluded and counted.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



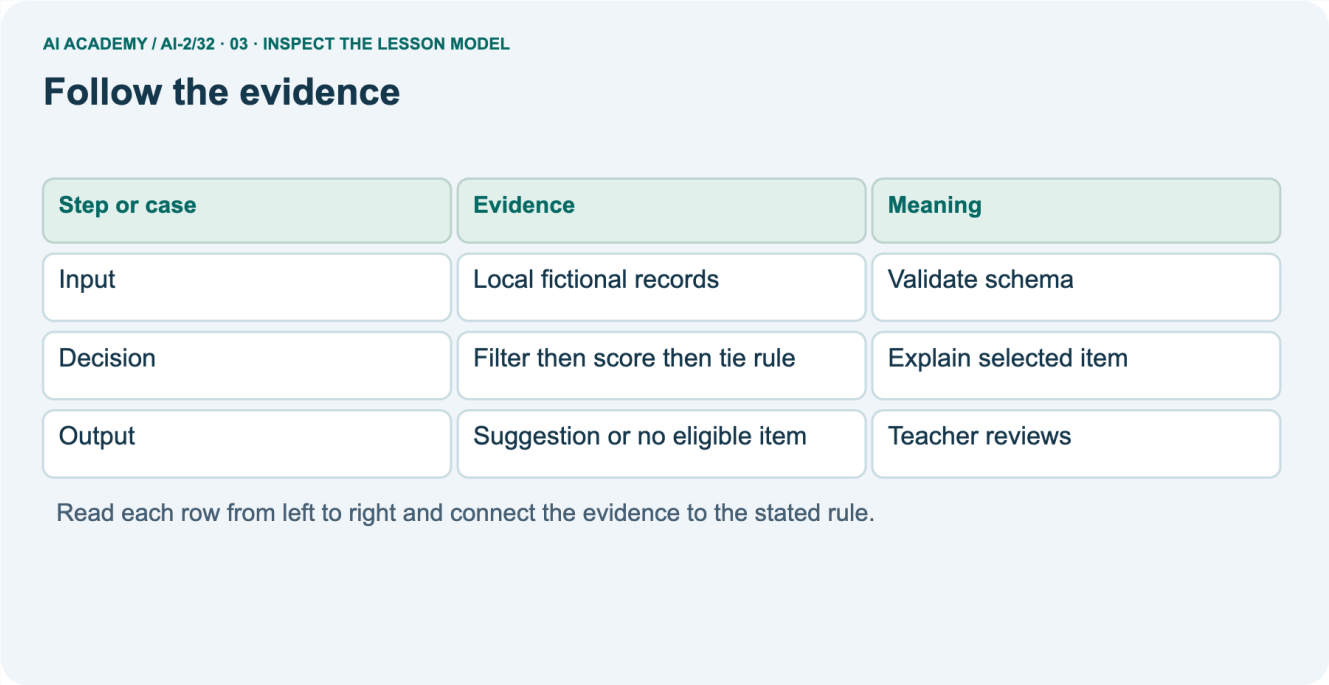
Follow the labelled picture with the matching explanation. Open the vector diagram

Specify **input**s: Validate each row and request; keep rejected-row reasons. → Filter and rank: At limit 8: A(time8,interest3), B(9,5), C(6,3).

B is ineligible. → Expected **output**: A beats tied C by ID; report A and its score 3. → **Acceptance case**s: Also plan invalid request, no eligible item and equal-score cases.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Follow the evidence

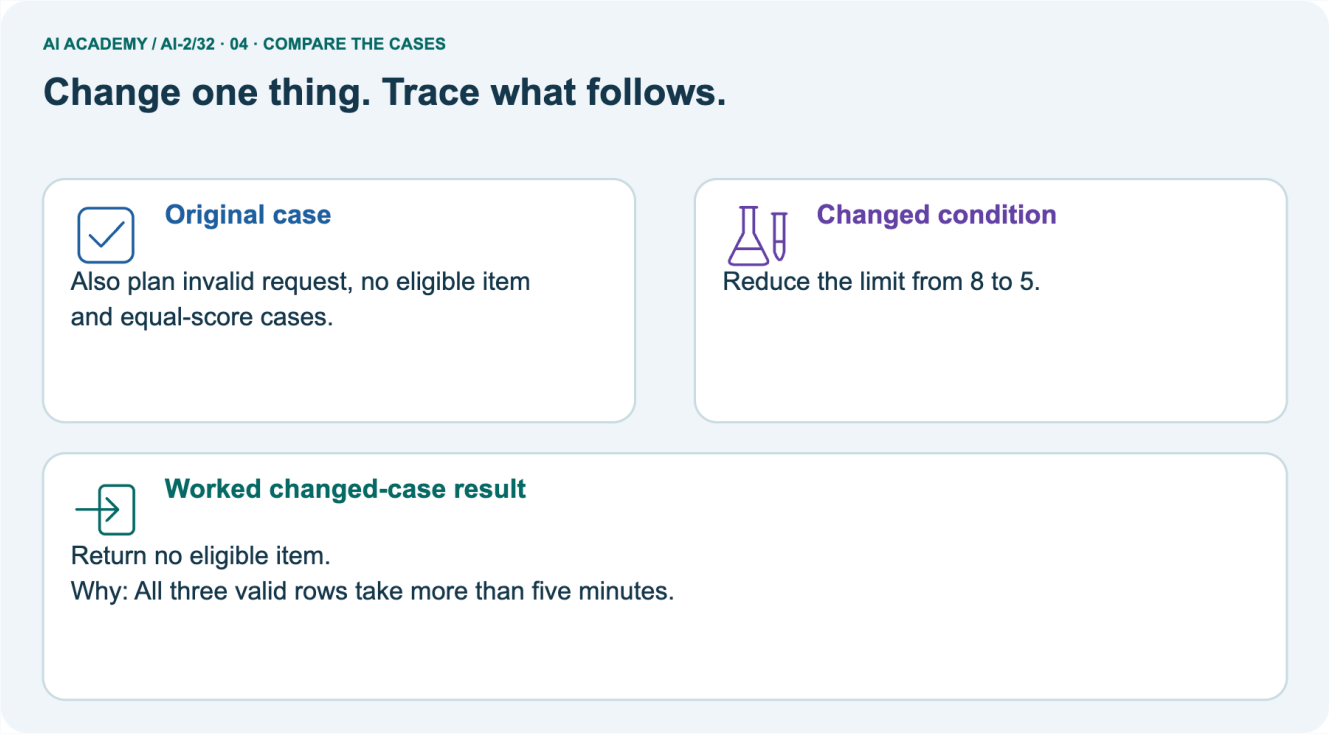


Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. **Input** | Local fictional records | Validate schema. Decision | Filter then score then tie rule | Explain selected item. **Output** | Suggestion or no eligible item | Teacher reviews

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



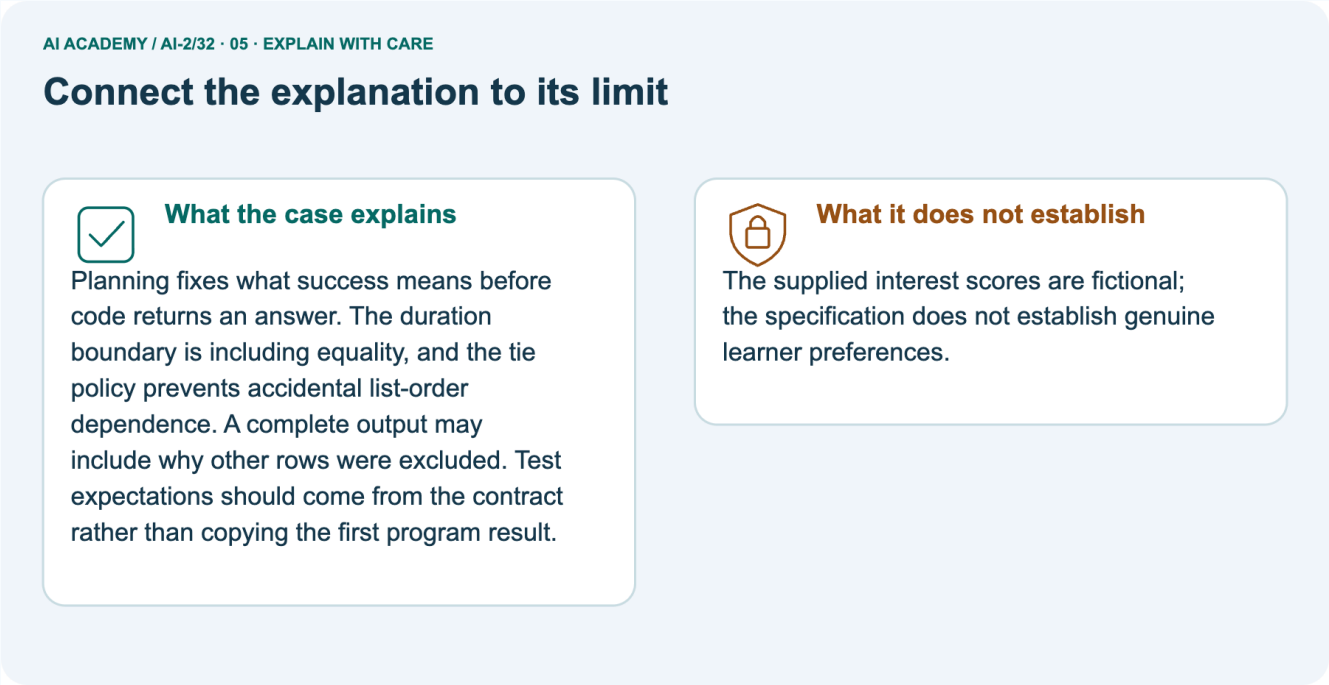
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Also plan invalid request, no eligible item and equal-score cases. Changed condition: Reduce the limit from 8 to 5. Expected result: Return no eligible item.

Reason: All three valid rows take more than five minutes.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

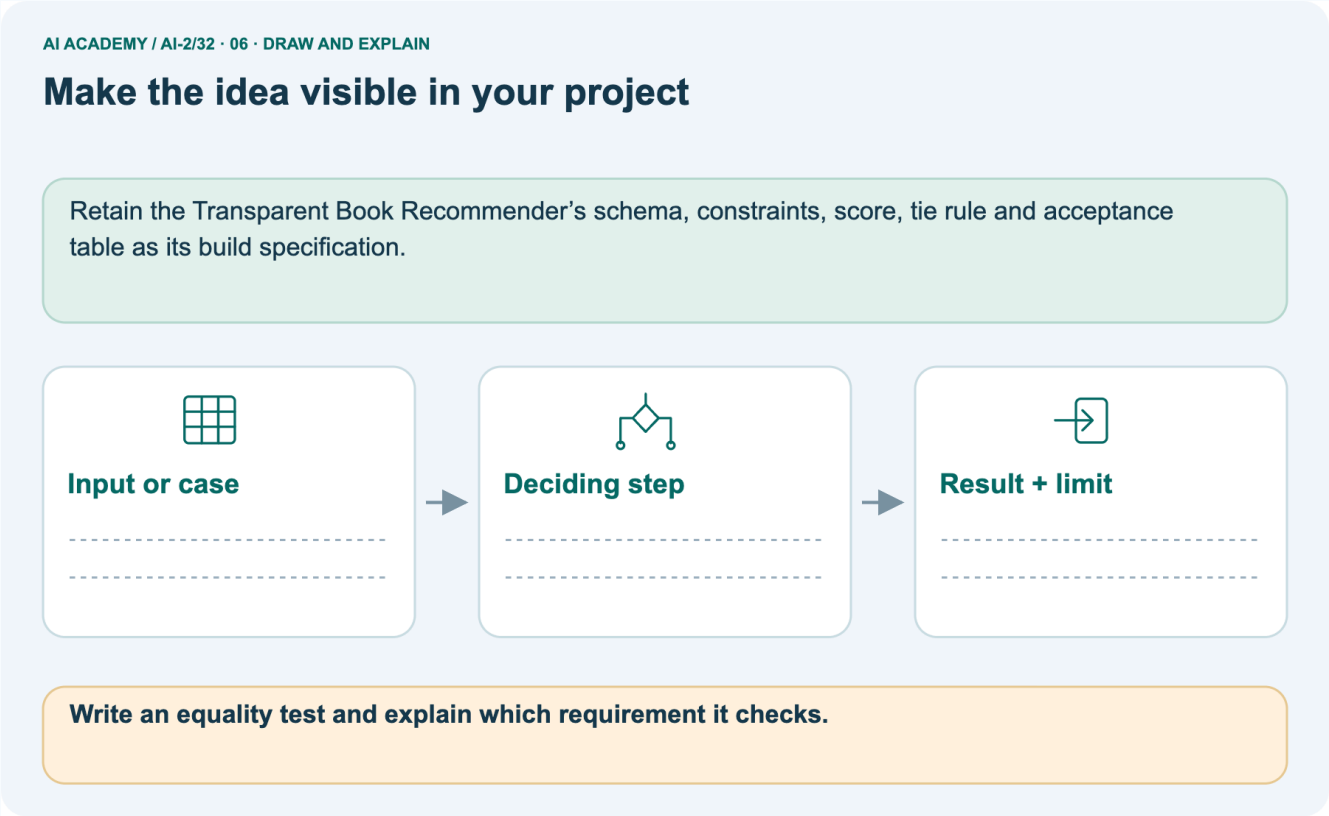
Planning fixes what success means before code returns an answer. The duration boundary is including equality, and the tie policy prevents accidental list-order dependence.

A complete **output** may include why other rows were excluded. Test expectations should come from the contract rather than copying the first program result.

Limit: The supplied interest scores are fictional; the specification does not establish genuine learner preferences.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Retain the Transparent Book Recommender's schema, constraints, score, tie rule and acceptance table as its build specification. Draw your **input**, deciding step, result and limit.

Supported question: Write an equality test and explain which **requirement** it checks.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.